

Phantom Wall: The Drone Dilemma

Project Description: This project explores the Drone Dilemma identified by Dr. Lin from Oak Ridge National Laboratory. The core issue is that unstable autonomous drones pose significant threats to public safety, including documented reports of drones colliding with oncoming traffic. This raised the question Dr Lin. posed, '**How do we safely ground these drone?**'.

The Phantom Wall introduces a non-kinetic system that neutralizes autonomous drones that operate without GPS or radar. The system is an innovative approach that advances from the jamming method to a logical sensor spoof method. It creates a hydro-projection of a fake wall to exploit the drone's obstacle avoidance sensors, like LiDAR, and trigger the 'stop-and-hover' protocol of the drone. This will then force the drone's battery to deplete to around 10%, forcing the drone to slowly descend to the ground below.

This project proposal, contains scope of research, a web-based simulation of the 'Phantom Wall', an annotated bibliography highlighting important sources that contributed to the success of this research project, and more.

Objectives

Build a reliable system to safely land drones that operate without GPS/radio signals, but use cameras and sensors.

Exploit drone's safety system by generating an adversarial visual pattern to trigger the drone's collision-avoidance safety protocol.

Ensure zero kinetic impact and prevent uncontrolled descent, to minimize civilian risk and collateral damage

Implement a real-time AI object tracker for optimal projection that is smart and fast.

Explore necessary protocol for extreme weather conditions and sunlight

Research Scope

Fully autonomous drones that do not use GPS or radar.

Methodology/Approach

Due to heavy science nature of this project, it is very important to research. Majority of the days I spent toward this project was research alone.

All sources used are found at the end of this document, in the 'Annotated Bibliography' section

Stakeholder

Catered to the Quickstart Internship at METIL lab, and more specifically catered to Oak Ridge National Laboratory and its representative, Dr. Lin.

Phantom Wall: The Drone Dilemma

Timeline

Phase I (December 11-14)

Proposed Plan:

Proposal Finalization

● **Feedback**

Refine proposal based on feedback given by Mr. Eakins

● **Strengthen technical detail**

The science, technology, AI, and fluid dynamics behind it

● **Expand on Possible Risks**

Daylight and tech limits

● **More Research**

Keep building the sources

Deliverables

Proposal Draft

Annotated Source List

Min Deliverables

Finalized problem/solution

Clear technical concepts that concern the project

What I achieved

I finalized the proposal based on the feedback provided by Mr. Eakins. I strengthened the technical detail, the science, technology, AI, and fluid dynamics behind it. This includes VSLAM, Theory of Computer Vision, Optical Flow, Laminar Flow, Reynolds Number, Adversarial Patterns and Denial of Service. I visited many websites, sources, and academic papers that studied these topics so I could get a better understanding. These sources are reflected in my Annotated Bibliography, at the end of this document. I explored possible risks like Daylight by researching Optical Physics and the size of water particles required to be visible to the sensors.

Phase II (December 14-19)

Proposed Plan:

Simulation & Tech Stack Finalization

● **Simulation Environment Prep**

Software for visual concepts, either Unity/Blender or 2D diagrams and figures

● **Input Data**

Research sample drone data (velocity, alt, etc.)

● **Define Methodology**

Step-by-step process

Deliverables

Simulation step-by-step documentation

System Architecture for Visuals

Min Deliverables

Needed Tech and Software Tools

Logic diagrams for the AI

What I achieved

During this time, I began implementing a browser based simulation. I chose to code it via VS Code rather than a simulation platform like Unity or Unreal so I would not spend more time on learning the tool but rather spend time conducting research, building sources, and creating a strong implementation.

Although not part of the plan, I continued researching the science and technology required to pull off the successful logic spoof system. I also wanted the system to work during any weather conditions and sunlight so I explored that as well.

I also drafted some use cases as advised by Mr. Eakins and my proposed plan for this phase. I did draft some necessary tools but I felt I was not able to construct a methodical and reasonable tool stack because I felt some of the research was lacking.

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Phase III (December 19-25)

Proposed Plan:

Design Doc & Visuals

Document tech stacks

Consolidate Methodology

System architecture

Visual Prototyping

Combine and Finalize Design Doc

Deliverables

Detail Schematics and Materials Needed

Visual Prototypes and Rendering

Min Deliverables

3D infographic or 2D infographic demonstration

Final Summary of Proposal

What I achieved

During this time I actually was sick for this entire phase with a fever and a cold. It was not however an excuse to not continue with the research and fulfilling this project. I just wore thick clothes, took a lot of advil, sweated out the fever multiple times, and focused on research and fixing my simulation.

Based on my proposed plan, I had already consolidated my technical stack, so I did begin composing my design document.

I also continued doing more research, but at this time I was pretty confident on the *Phantom Wall's* ability to work.

Phase IV (December 25 - January 2)

Proposed Plan:

Literature Review and Finalization

Mark Citations

Proofread

Feedback from Others

Deliverables

Polished Literature Review

Min Deliverables

Citation List

What I achieved

During the start I was still recovering from my fever and sickness, but when I finally recovered on December 28th, I went back into the normal project routine.

I finalized my implementation/simulation by debugging the code. During this time, I uploaded the code that I had as a .html file in Notepad ++, and ran the code on my own computer, and launched it on google to run it.

For my citation and Literature Review, I consolidated and polished them. I had most of the sources already listed, but I had not chosen my critical sources up until this point. I also detailed the significance of each critical source in the Literature Review.

I did show a peer member what I had, but I did not receive much feedback because at that point in time there was not much to critique.

During this time I was also doing more and more research as I had some doubts about the ability to conduct the system in the sunlight so I instead focused on a nighttime focused system.

Towards the end, I ended I had a finalized Design document, proofread it, asked my peers for their advice, then submitted it on January 2nd.

Phantom Wall: The Drone Dilemma

System Architecture 3000 gallons of water This can be filled in a take. It is needed to operate up to 13 minutes. 50kW diesel and generator This is needed to power the projector and the AI logic detection. A projector of at least 30000 lumens This is necessary to create the illusion of the wall. Dual-Layered System Layer I This first layer fools the camera. It maintains laminar-flow to create a the visual surface to project adversarial images. Layer II The layer behind Layer I acts as defense system against the drone's depth perception sensors. There is a perimeter of high-pressure mist nozzles that generate a dense cloud of fog surrounding the screen. It also scatters LiDAR and obscures the "edges" of the projection, to create a barrier that appears infinite and solid to depth sensors. High Water Pressure and Heavy Nozzle Weights This is needed to support low winds, lower than 15 mph.	Specifications Must be operated at night Due to the systems light and holographic nature, it is necessary to only operate the <i>Phantom Wall</i>, at night. This is because the sunlight at sea level provides 100,000 lux of luminance, while ambient light contributes between 10,000 and 25,000 lux. Computer Vision sensors operate on contrast, similar to human vision. It uses relative difference in luminance between an object and its background. To make the projector image visible to the sensors, the display must achieve a contrast ratio of at least 5:1 against its background. For the <i>Phantom Wall</i> to be apparent and captured by the drone's sensors, the projector light would need to be multiplied by a factor of 5 , which technology infeasible. Therefore, it is required to operate the Phantom Wall at night.
Real Time AI Surface Tracking This is known as MIDAS, or Markerless and Modeless Dynamic Projection Mapping An AI image warping system is necessary so that the projected static image does not appear warped or distorted. When the distorted image hits distorted water, the two distortions create a perfectly flat and stable image for the drone. Specifications GPU (like NVIDIA Jetson Orion or RTX 6000 Ada) is needed for processing power. High-speed infrared (IR) depth camera that operates 500 fps or more to scan the water surface is necessary.	Must be operated without high winds High speed winds up to 15–20 mph and beyond would 'break' the phantom wall, distorting the image significantly, and lose the 'solid-like' effect of the wall. Must be operated near an available water hydrant This is due to large amount of water necessary to carry out the <i>Phantom Wall</i>. Must not be operated during extreme weather conditions Similar to high winds, extreme weather conditions break the integrity of the <i>Phantom Wall</i>.

Use Case & Deployment

Mobile Deployment

Necessary technical stack

High-Pressure Pump up to 1500 GPM @ 150 PSI

Water Tank of 3,000 gallons

Generator powered by 50kW Diesel

30,000 Lumen Projector

Computer for the AI surface tracking and the adversarial pattern

Tracking Sensors

Needed Personnel

A **driver** to manage the vehicle's position.

A **pump operator**, to engage the pump and regulate the water pressure to maintain the screen's laminar flow.

A **system operator** to operate the AI targeting system, select the adversarial pattern projected, and monitor the drone's behavior using the tracking interface.

A **safety member** to monitor airspace and ground traffic and maintain the safety for civilians.

Scenario

An unstable drone is flying at 10m altitude along a busy highway, posing a safety concern to oncoming traffic. It is GPS denied but uses cameras to operate.

Phantom Wall Execution

1. Use the Mobile unit and enter the highway ahead of the drone
2. The mobile stops well ahead of the drone, and positions the Phantom wall in the drone's projected path.
3. The pump is engaged, a 10 by 6 meter wide laminar water screen is projected.
4. The AI tracker focuses on the drone, creates an adversarial pattern directly on the water screen, and limits image distortion.
5. The drone's collision avoidance detects the 'wall' and brakes and hovers. Its battery depletes to 10%, and slowly descends to the ground below.

Target Drone Specifications and Compatibility

Compatible With

DJI (Mavic, Matrice, Mini)

Highly compatible as it rely heavily on visual and LiDAR fusion

Skydio (2/X2/X10)

Compatible but can be challenging as the system is duped by curved walls rather than flat obstacles. An approach is to set up multiple Phantom Walls to box this drone in.

Autel (Evo)

Highly compatible, similar to DJI.

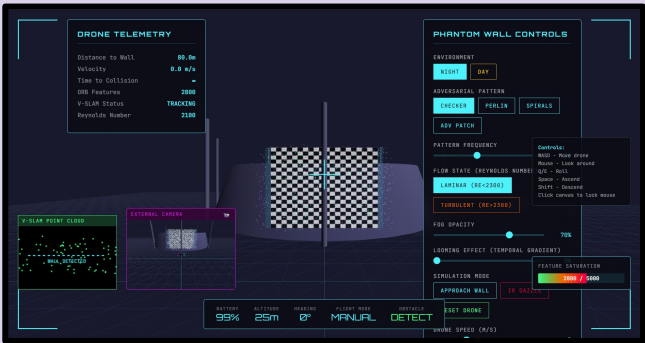
Not Compatible with

FPV Racing Drones, Pre-programmed inertial drones that are military grade, and Thermal-Only Navigation drones.

Simulated Implementation

Description: The Phantom Wall Simulation is a very simple implementation tool that simulates a drone approaching the virtual *Phantom Wall*. The user interface highlights the metrics of the drone itself, what it's sensors see, the main components for the virtual to be perceived, as well as the ability to adjust those components.

Phantom Wall Simulation Video Demo



Conclusion

I really enjoyed researching and constructing this project. There were some ideas I had that involved using more drones, almost like a rescue drone scenario but it seemed a bit counterintuitive to use more drones.

As the project relies heavily on complex optical physics it is definitely necessary to conduct live tests. The web based simulation and technical research is great for the initial approach but to actually legitimize the project, it is necessary to use real experimentations.

Overall, I believe this proposal provides a very adequate proposition for a non kinetic approach to ground unstable drones.

Nuances to Explore

Daytime Capability

Something I really worked hard on as part of the research side was making sure that this virtual wall would work during the daytime. It would take this project to a whole new level if it could function during the daytime. I believe, through more research and changing some of the technical specifications, it could be possible.

Live Testing

Through live testing, nuances can not only be explored but discovered. I believe testing is crucial, and a live environment offers more realistic insights than research and coding alone.

Challenges Faced and What I Learned

Research

I think the biggest challenge for me was the research portion just because this project is rooted so much in optical physics, the science of water particles and light.

Independency

Working from home meant I was my own manager. I ought to be on top of time management and responsible for my own actions that involved the project. It was a great experience because it showed me that I could be independent, and although it was challenging I learned how to divide my time efficiently.

Sickness

I am not sure what it was but it could have been something about the cold weather that gave me a cold and fever. My fever last 7 days so I spent most of my time bedridden, but I did not use it as an excuse to delay the project. Took advil daily and smothered myself with lots of layers and blankets until I sweated out the fever twice. After the 7 days I was left with a cold that I finally got rid of within 4 days.

Initial Idea: Solution First

I actually really wanted to have a VR based simulation that prioritized building teamworking skills. I was inspired by medical simulation by Vcom3D. However after discussing my idea the day after Simulation event, through some advice I realized I was aiming for a technology/solution first innovation rather than problem-then-solution.

Annotated Bibliography

"3D Holographic Projection Systems - Holographic Stage." Virtual On, virtualongroup.com/3d-holographic-projection-holographic-stage/. Accessed 8 Dec. 2025.

This page introduces the concept of 3D Holographic projection systems and contains a catalogue of onstage variations. It is the overall inspiration for the project proposal.

"Advexure." Advexure, 2025, advexure.com/collections/multispectral-drones-sensors. Accessed 9 Dec. 2026.

The page serves as an explanation of how multispectral drone sensors work. It also highlights different autonomous drones used commercially.

"AeroVironment Expands Puma™ Visual Navigation System (VNS) Kit to Puma LE | AeroVironment, Inc." AeroVironment, Inc., 2025, investor.avinc.com/news-releases/news-release-details/aerovironment-expands-pumatm-visual-navigation-system-vns-kit. Accessed 10 Dec. 2026.

AeroVironment shows how they integrate their 'Visual Navigation System' (VNS) kit with their Puma LE unmanned aircraft, to enable precise navigation where GPS signals have been jammed. This source highlights the advancements in technology to safeguard drones against jamming, showing how jamming is becoming outdated in warfare or conflicts.

EA-J100 Agricultural Spraying Drone User Manual V1.0 3WWDZ-45A Intelligent Quadrotor Agricultural Spraying Drone. 2023.

This is a user manual for a specific Agricultural Spraying EA-J100 drone, the manual offers insights on the specific obstacle avoidance programming in the system.

"High Bright Solutions (Sun Readable Monitors) - PRTronic GmbH." Prtronic.at, 2026, www.prtronic.at/High_bright_solutions/. Accessed 10 Dec. 2026.

The page explores invisibility of holographic and projections in a high ambient sunlight environment. It shows how it is ineffective during daytime use because of 5:1 contrast requirement. It supports my project's 'only operate during night time' claim.

Kim, Jonguk, et al. "Image Distortion and Rectification Calibration Algorithms and Validation Technique for a Stereo Camera." *Electronics*, vol. 10, no. 3, 1 Feb. 2021, p. 339, <https://doi.org/10.3390/electronics10030339>. Accessed 19 Dec. 2025.

This academic paper highlights image rectification and calibration algorithms that are used by stereo-vision systems to calculate detect obstacles in the absence of GPS. It provides the technical basis for identifying vulnerabilities when manipulated visual inputs can disrupt the drone's spatial analysis and allow for exploitation of its obstacle avoidance mechanisms.

Murefu, Freddy. "Ways AI Is Changing Autonomous Drone Navigation." *AI Wakforce*, 28 Apr. 2024,

www.aiwakforce.com/blog/the-astounding-way-ai-is-changing-autonomous-drone-navigation/. Accessed 11 Dec. 2026.

This blog serves as evidence of current obsolete and outdated jamming methods. It supports how the project proposal 'fills in research-gaps'.

Pavlo Kryvenko. "Ukrainian Drones vs Russian Jamming - New Geopolitics Research Network." *New Geopolitics Research Network*, 10 June 2024, www.newgeopolitics.org/2024/06/10/ukrainian-drones-vs-russian-jamming/. Accessed 5 Dec. 2025.

This online article compares Ukrainian drones and russian jamming in 2024. It highlights the how in the face of Russian jamming, Ukrainian engineers and developers started to engineer innovated solutions that relied only on sensors. This is to bypass bypassing the need for GPS and other signals susceptible to jamming.